



Convergent Privacy Framework for Multi-layer GNNs through Contractive Message Passing



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Summary of Contribution



- ◆ A **novel privacy analysis** for GNNs that leverages the **contractiveness** of message passing to achieve the **convergent** privacy costs.
- ◆ **New design** of *perturbed graph contractive layer* and a practical private GNN **framework CARIBOU**.



Full-version Paper

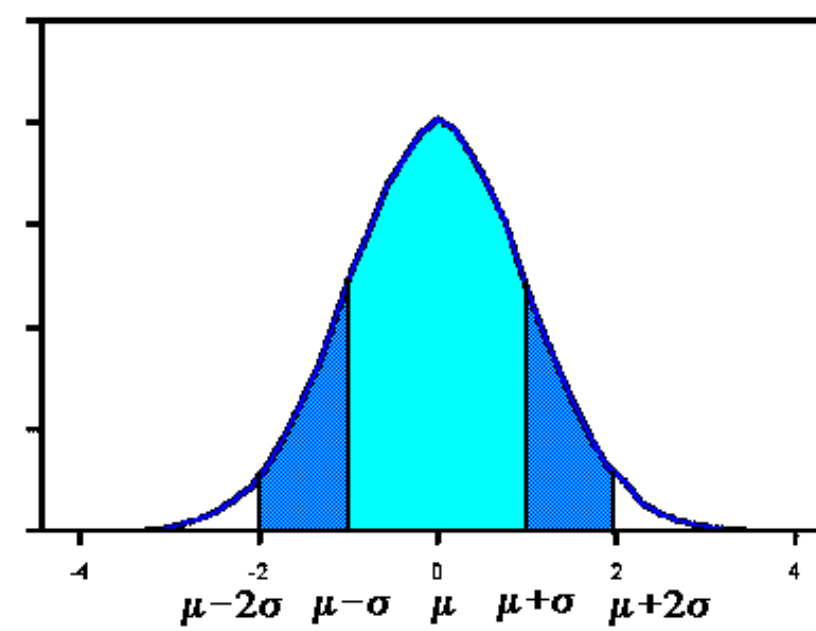
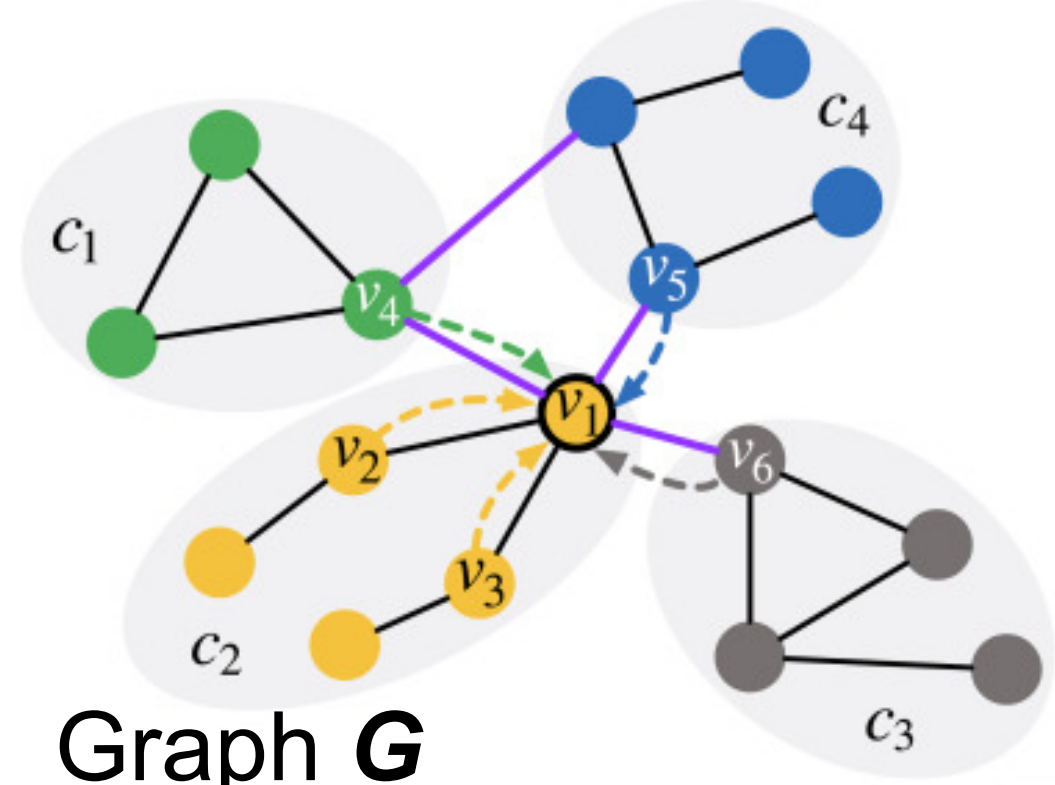


Code



Scenarios & Motivation

Perturbed Message Passing with Differential Privacy (DP)



G, G' : Edge/node-level neighboring.

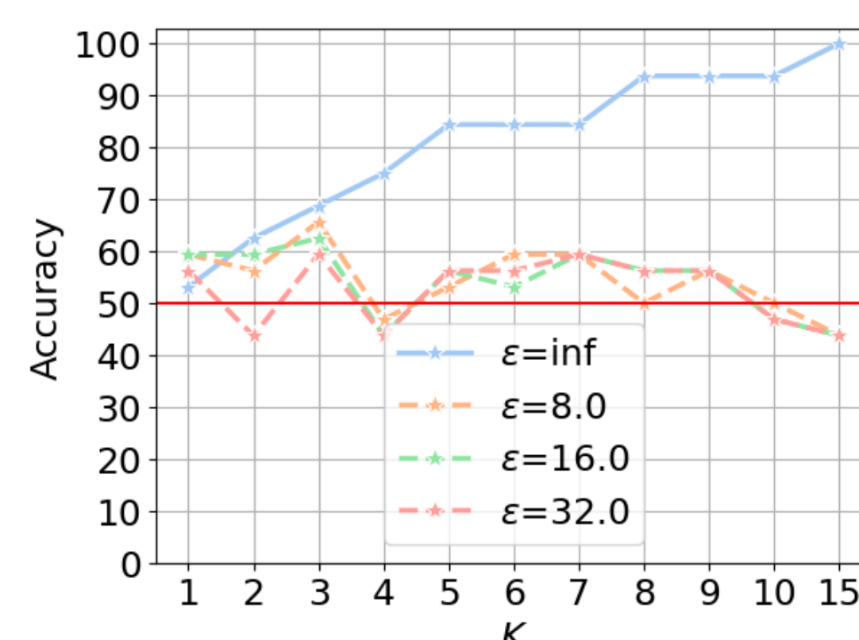
$$\text{Formulation: } \mathbf{X}^{(k+1)} = \Pi(\text{MP}_G(\mathbf{X}^{(k)}) + \mathbf{Z}^{(k)})$$

Node representation/embeddings from message passing (MP); $\mathbf{X}^{(0)}$: input feature matrix.

Gaussian noise $\sim N(0, \sigma^2)$.

Framework	Mechanism	Noise (σ^2)	Utility
PertGraph [1,2]	G Pert.	$\propto 1$	Fair
DPDGC [3]	Decoupled G w/ Pert.	$\propto K$	Good
GAP [4]	Pert. MP	$\propto K$	Good

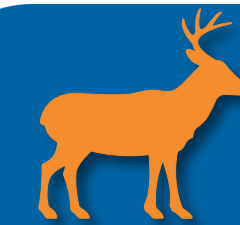
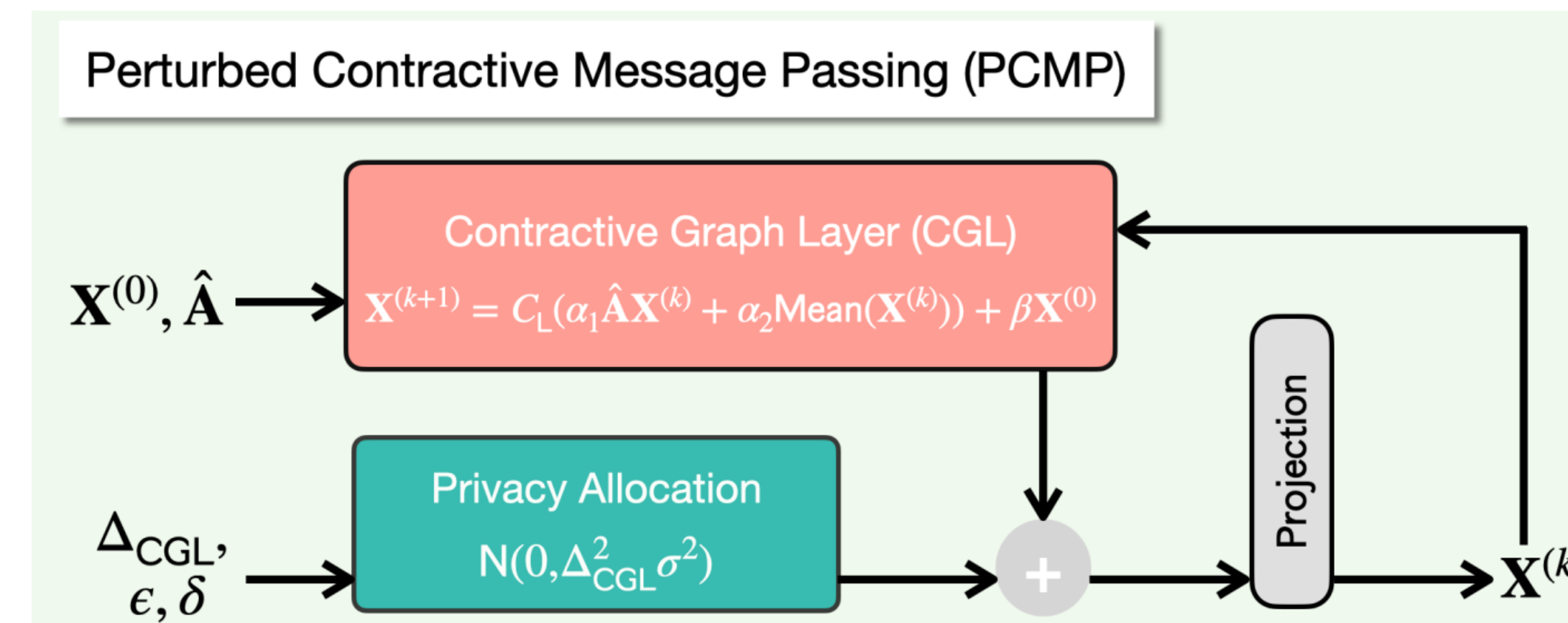
- ◆ Share a **critical limitation**: privacy loss **grows linearly with K** .
- ◆ Require large σ to maintain a reasonable privacy level for multi-layer GNNs, **degrading utility**.



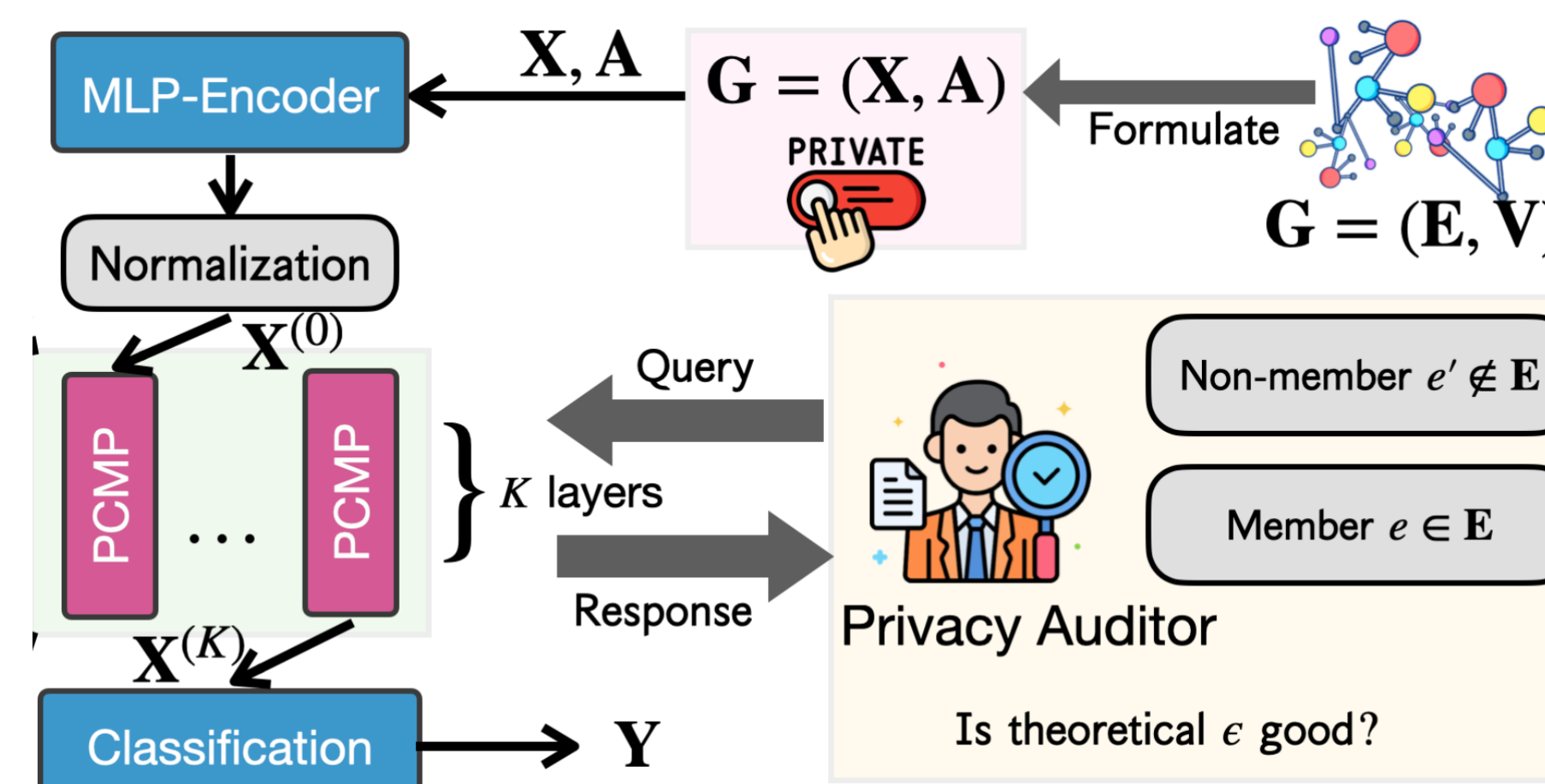
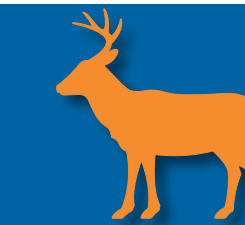
Convergent Privacy Budget



- ◆ **Insight**: leverage the inherent **privacy amplification** that occurs in multi-layer GNNs through **contractiveness**.
 - ❖ When perturbed MP is contractive, the distance between GNNs trained on neighboring G, G' shrinks at each step.
 - ❖ Consequently, the influence of individual data points diminishes, leading to the amplified privacy rooted from “over-smoothing.”
 - ✓ **Remove the over-estimated privacy loss.**
- ◆ Derive a much **tighter bound** for the **finally released** GNN model.



CARIBOU Framework



- ◆ **Theorem 1 [DP guarantee for CGL layers]**. Let G be a graph and K be the number of contractive graph layers in CARIBOU. Let $C_L < 1$ be Lipschitz constant. Then, the K -hop message passing of CARIBOU satisfies: $(\frac{\alpha \Delta^2}{2 \sigma^2} \min\{K, \frac{1 - C_L^K}{1 + C_L^K} \frac{1 + C_L}{1 - C_L} + \frac{\log(1/\delta)}{\alpha - 1}\}, \delta)$ -DP, where σ is the noise scale, Δ is the sensitivity of MP, and α, δ are DP parameters.



Experiments

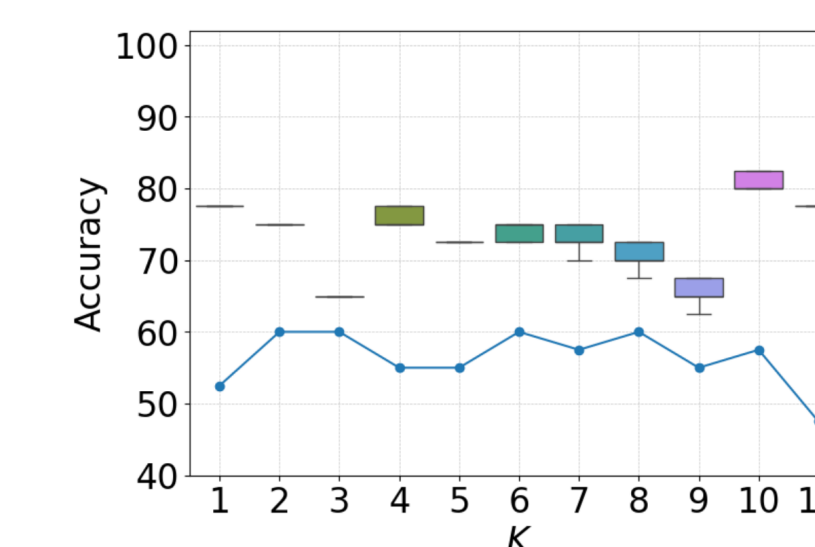


Accuracy: EDP over the Cora dataset.

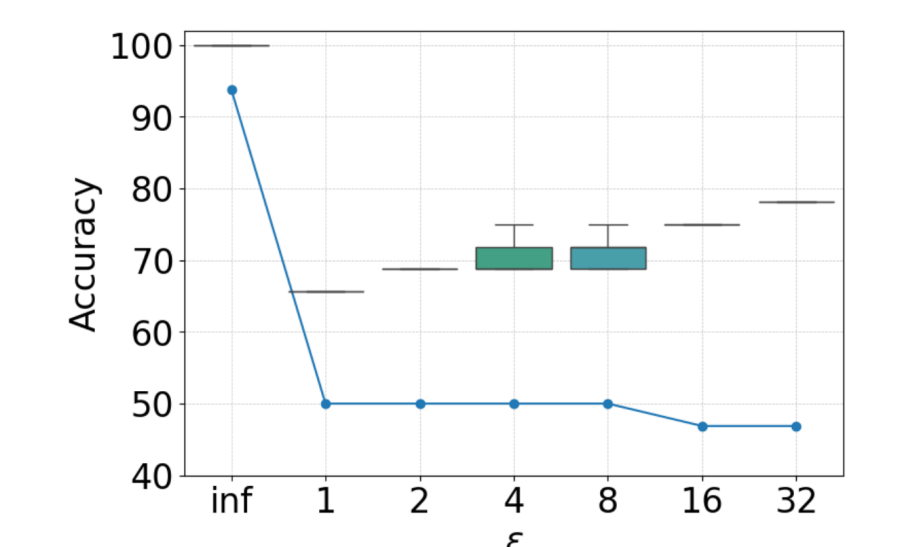
ϵ	1	2	4	8	16	32
CARIBOU	85%	87%	87%	88%	89%	89%
GAP	76%	78%	75%	76%	78%	80%

Accuracy: NDP over the Cora dataset.

ϵ	1	2	4	8	16	32
CARIBOU	81%	83%	86%	87%	88%	88%
GAP	34%	32%	32%	44%	56%	64%

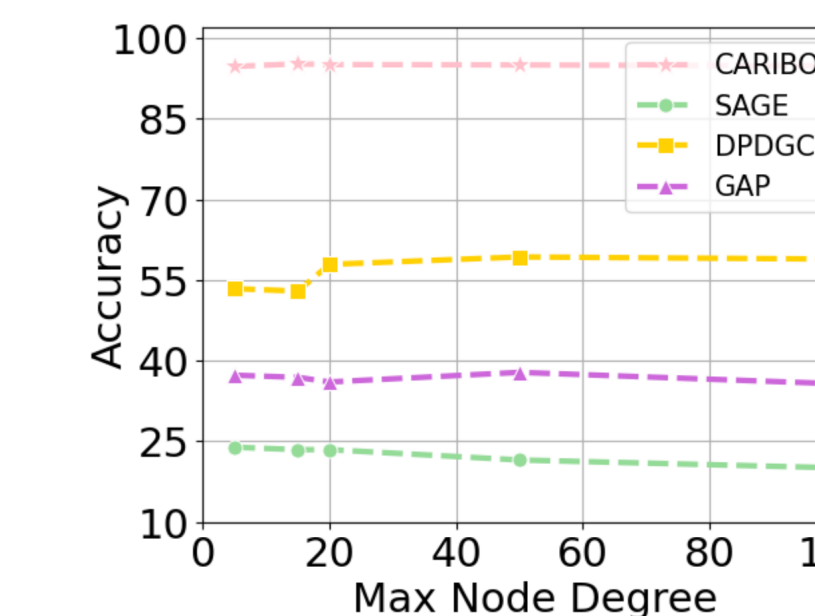


(a) Different K when $\epsilon = 4$

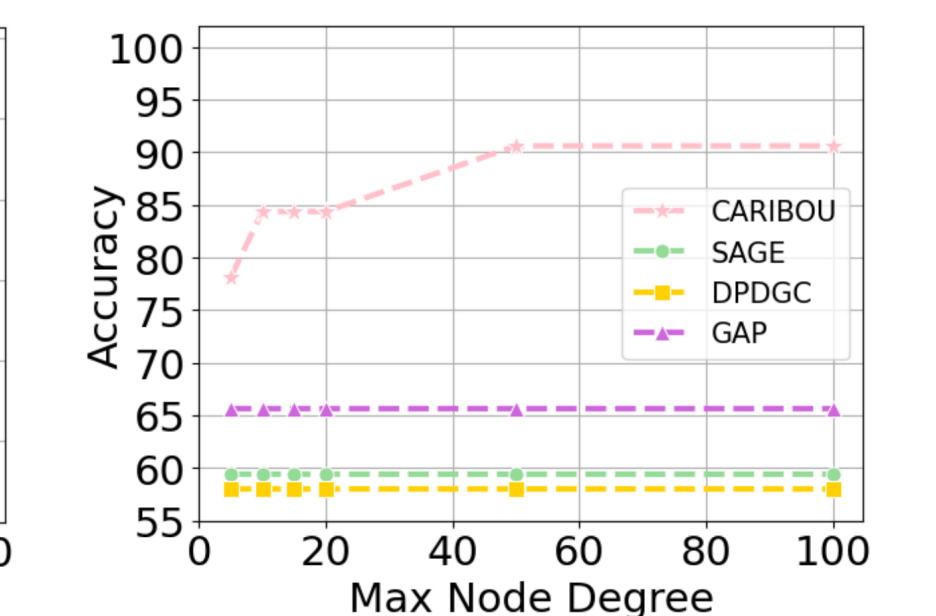


(b) Different ϵ when $K = 10$

Fig. 5: Comparison between CARIBOU (colored boxes) and GAP (blue lines) for ablation study.



(a) $\epsilon = 2$ (Photo)



(b) $\epsilon = 2$ (Chain-S)

Fig. 6: NDP Accuracy with Varying Max Node Degree.

Acknowledge & Reference

- [1] A. Kolluri, T. Baluta, B. Hooi, and P. Saxena, “Lpgnet: Link private graph networks for node classification,” in ACM SIGSAC Conference on Computer and Communications Security, CCS, 2022, pp. 1813–1827.
- [2] F. Wu, Y. Long, C. Zhang, and B. Li, “LINKTELLER: recovering private edges from graph neural networks via influence analysis,” in IEEE Symposium on Security and Privacy, SP, 2022, pp. 2005–2024.
- [3] E. Chien, W.-N. Chen, C. Pan, P. Li, A. Ozgur, and O. Milenkovic, “Differentially private decoupled graph convolutions for multigranular topology protection,” in Advances in Neural Information Processing Systems, vol. 36, 2023.
- [4] S. Sajadmanesh, A. S. Shamsabadi, A. Bellet, and D. Gatica-Perez, “Gap: Differentially private graph neural networks with aggregation perturbation,” in USENIX Security 2023-32nd USENIX Security Symposium, 2023.

